Ms. Welles **Alg 2 Trig/Honors**

<u>Chp 6.4: More Trigonometric Graphs</u>

I. Graphs of Tangent, Cotangent, Secant, and Cosecant

Periodic Properties:

The functions tangent and cotangent have period π.

$$\boxed{\tan(x+\pi) = \tan x \qquad \cot(x+\pi) = \cot x}$$

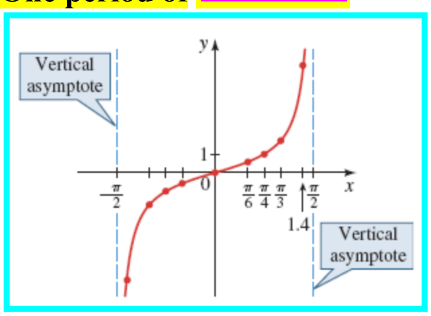
The functions cosecant and secant have period 2π.

$$\boxed{\csc(x+2\pi) = \csc x \qquad \sec(x+2\pi) = \sec x}$$

<u>TANGENT</u>

The graph of $y = tan\ x$ approaches the vertical lines $x = \frac{\pi}{2}$ and $x = -\frac{\pi}{2}$, so these lines are vertical asymptotes. The vertical asymptotes of $y = tanx$ are $x = \frac{\pi}{2} + n\pi$ where n is an integer.

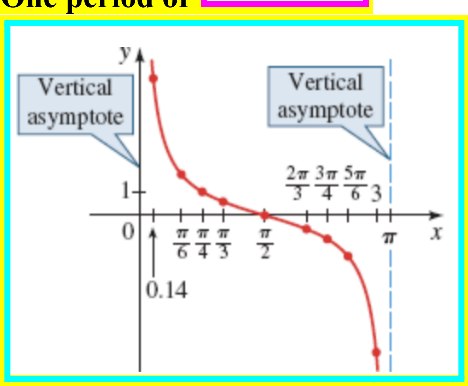
One period of $\boxed{y = \tan x}$



<u>COTANGENT</u>

The graph of $y = cot\ x$ is undefined for $x = n\pi$, with n an integer, so its graph has vertical asymptotes at these values. The vertical asymptotes of $y = cot\ x$ are $x = n\pi$ where n is an integer.

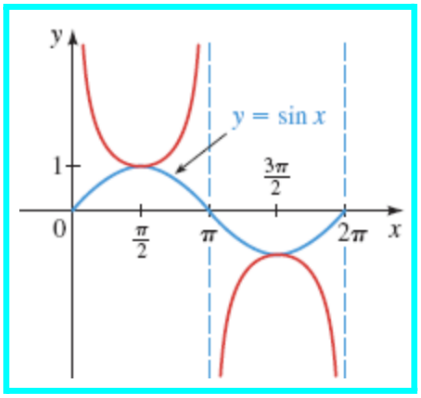
One period of $\boxed{y = \cot x}$



COSECANT

To graph the cosecant we use the identity $csc\ x = \frac{1}{sin\ x}$. The graph of $y = csc\ x$ has vertical asymptotes at $x = n\pi$, for n an integer.

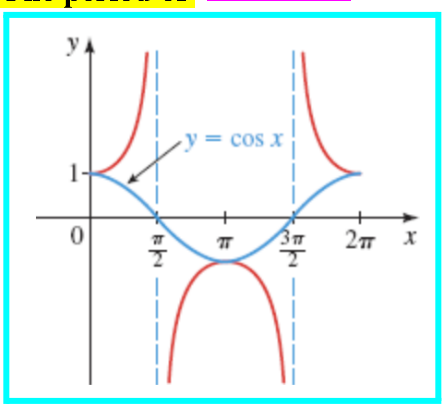
One period of $y = \csc x$



SECANT

To graph the secant we use the identity $sec\ x = \frac{1}{cos\ x}$. The graph of $y = sec\ x$ has vertical asymptotes at $x = (\pi/2) + n\pi$, for n an integer

One period of $y = \sec x$



Graphs of Transformations of Tangent and Cotangent

The functions $y = a\ tan\ kx$ and $y = a\ cot\ kx$, $(k > 0)$, have period π/k.

To graph one period of $y = a\ tan\ kx$, an appropriate interval is $\left(-\frac{\pi}{2k}, \frac{\pi}{2k}\right)$.

To graph one period of $y = a\ tan\ k(x - b)$, an appropriate interval is $\left(-\frac{\pi}{2k} + b, \frac{\pi}{2k} + b\right)$

To graph one period of $y = a\ cot\ kx$, an appropriate interval is $\left(0, \frac{\pi}{k}\right)$.

To graph one period of $y = a\ cot\ k(x - b)$, an appropriate interval is $\left(0 + b, \frac{\pi}{k} + b\right)$

II. Graphs of Transformations of Cosecant and Secant

The functions $y = a\ csc\ kx$ and $y = a\ sec\ kx$, $(k > 0)$, have period $2\pi/k$.

An appropriate interval on which to graph one complete period is $[0, 2\pi/k]$.

To graph one period of $y = a\ csc\ k(x - b)$ or $y = a\ sec\ k(x - b)$, an appropriate interval is $\left(0 + b, \frac{2\pi}{k} + b\right)$

Example 1: Find the period, and graph the function.

a. $y = 4tan(4x - 2\pi)$

b. $y = 2csc\left(\pi x - \frac{\pi}{3}\right)$

c. $y = 5\,sec\left(3x - \frac{\pi}{2}\right)$

d. $y = 2\,cot\,(3\pi x + 3\pi)$